ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL SCIENCES P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Agricultural Sciences Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept other relevant, correct answers.

SECTION A

1.1 1.1.1 C ✓✓ 1.1.2 A ✓✓ 1.1.3 B ✓✓ 1.1.4 A ✓✓ 1.1.5 C ✓✓ (10)

1.2 1.2.1 Weathering ✓✓ 1.2.2 Humus ✓✓ 1.2.3 Soil texture ✓✓ 1.2.4 Biome ✓✓ 1.2.5 Transpiration ✓✓ (10)

1.3 1.3.1 Clay ✓✓ 1.3.2 Sulphur ✓✓ 1.3.3 carbon dioxide ✓✓ 1.3.4 Bacteria / fungi (micro-organisms) ✓✓ 1.3.5 nitrogen ✓✓ (10)

TOTAL SECTION A: 30

SECTION B

QUESTION 2

2.1 Parent material ✓; climate ✓; living organisms ✓; topography (relief) ✓; time ✓ (5)

2.2.1 Sample B ✓. It has a balanced mixture of sand, silt and clay ✓. (2)

2.2.2 Sample A ✓. It has the most sand (80%) ✓; sand particles are large with big pore spaces, so water drains through quickly ✓. (3)

2.2.3 Sample C ✓. Its pH of 5,2 is acidic, and lime raises the pH ✓. (2)

2.2.4 Any TWO: poor drainage / waterlogging ✓; hard to work (sticky when wet, hard when dry) ✓; poor aeration; acidity reduces nutrient availability (2)

2.3 Any FOUR: improves soil structure ✓; increases water-holding capacity ✓; adds plant nutrients ✓; improves aeration and drainage ✓; feeds soil organisms; reduces erosion; buffers soil pH (4)

2.4 Any THREE: contour ploughing ✓; cover crops / keep the soil covered ✓; terraces ✓; windbreaks; avoid overgrazing; mulching; minimum tillage (3)

2.5 Compaction reduces the pore spaces in the soil ✓, so less air and water can enter ✓. Roots struggle to grow through the hard soil ✓, so plants take up less water and nutrients and grow poorly ✓. (4)

[25]

QUESTION 3

3.1.1 Producer: grass ✓. Primary consumer: cattle ✓. (2)

3.1.2 Most energy is used by the organism for life processes such as respiration and movement ✓, some is lost as heat ✓, and some is in parts that are not eaten or digested (lost in waste) ✓. (3)

3.2 Biotic factors are the living parts of an ecosystem ✓, e.g. plants, animals, micro-organisms ✓✓. Abiotic factors are the non-living parts ✓, e.g. temperature, rainfall, soil, light ✓✓. (6)

3.3 Any TWO: grassland ✓; savanna ✓; Nama Karoo (for small stock). Reason for one, e.g. grassland has plenty of grass for grazing ✓. (3)

3.4 Overgrazing: too many animals eat the grass faster than it can regrow ✓; the soil is left bare ✓ and erodes ✓; good grasses are replaced by weeds and bush (bush encroachment) ✓. Prevention, any TWO: keep the correct number of animals for the land (carrying capacity) ✓; rotational grazing / resting camps ✓; provide extra feed in dry seasons. (6)

[20]

QUESTION 4

4.1 Carbon dioxide + water ✓ → (light energy, chlorophyll) ✓ → glucose + oxygen ✓ (3)

4.2 Any THREE: light intensity ✓; carbon dioxide concentration ✓; temperature ✓; availability of water; amount of chlorophyll (3)

4.3 Any TWO, explained: the temperature is kept warm and stable, so enzymes work well ✓✓; carbon dioxide levels can be increased ✓✓; water can be controlled through irrigation; plants are protected from wind and frost. (4)

4.4 (5)

Photosynthesis	Respiration
Takes place in chloroplasts	Takes place in mitochondria ✓✓
Uses CO_2 and water; releases oxygen	Uses glucose and oxygen; releases CO_2 ✓✓
Stores energy (makes food)	Releases energy (breaks down food)

Any TWO pairs; table with headings ✓

4.5.1 Nitrogen (N) ✓; phosphorus (P) ✓; potassium (K) ✓ (3)

4.5.2 22% of the mass of the bag is made up of these nutrients ✓✓ (the total nutrient content). (2)

4.5.3 $\text{Nitrogen} = \frac{2}{7} \times 22\% = 6{,}29\%$ ✓; $6{,}29\% \times 50\ \text{kg}$ ✓ $= 3{,}14\ \text{kg}$ ✓ (3)

4.6 Advantage, any ONE: improves soil structure / cheap or free / adds organic matter ✓. Disadvantage, any ONE: low and uncertain nutrient content / bulky to transport / may contain weed seeds ✓. (2)

[25]

TOTAL: 100

END OF MARKING GUIDELINE